LECTURE 07 · REVIEW NOTES

Fixed Income Markets

How bonds are priced, what drives yields, and how investors measure and trade credit risk.

About these notes

These notes review the material of Lecture 07. Each concept is explained in full, and each numerical example is worked out step by step with every intermediate figure shown. Exercises 1 and 2 are reproduced with complete solutions. Relevant readings: Fabozzi, *Fixed Income Mathematics*, Chapters 1–4; Brealey, Myers & Allen, Chapter 3.

1. What is a Bond?

A bond is a loan packaged as a tradeable security. The issuer — a government, municipality, or corporation — borrows a fixed sum from investors and, in return, commits to paying a series of periodic interest payments called **coupons** and repaying the **face value** (or par value) on a specified maturity date. In the United States the standard convention is semi-annual coupons, so a bond with a 6% annual coupon and a $1,000 face value pays $30 every six months.

Four numbers define any bond: the **face value** (the amount repaid at maturity, conventionally $1,000 or $100); the **coupon rate** (annual interest as a percentage of face value); the **maturity** (the number of years until the face value is repaid, ranging from weeks for T-bills to 30 years for long bonds); and the **price** (what an investor pays today in the secondary market, which is rarely equal to face value). The price–face value relationship is informative: when price equals face value the bond trades **at par**; above face value it trades at a **premium**; below face value, at a **discount**.

The Fixed Income Market in Context

The global fixed income market is substantially larger than the global equity market — roughly $130 trillion outstanding in 2024 versus approximately $110 trillion in equities. The United States alone accounts for more than $50 trillion, making it the world's largest single bond market. Understanding fixed income is therefore not a narrow specialty: it is central to how governments fund themselves, how corporations raise capital, and how monetary policy is transmitted through the economy.

Issuer Type	Examples	Approx. Share
Governments	US Treasuries, Bunds, JGBs	~40%
Municipalities	State & city bonds (often tax-exempt)	~5%
Corporations	Investment-grade & high-yield	~25%
Supranationals / other	World Bank, IMF, agency MBS	~30%

2. Yield to Maturity

The **yield to maturity (YTM)** is the single discount rate that equates the present value of all future cash flows — coupons plus the face value at maturity — to the bond's current price. It is the fixed income analogue of an internal rate of return (IRR): if you buy the bond today at the market price and hold it to maturity, reinvesting every coupon at the same rate, the annualized return you earn is exactly the YTM.

The YTM formula for a semi-annual coupon bond is:

$$\text{Price} = \sum_{t=1}^{2n} \left[\frac{C/2}{(1 + \text{YTM}/2)^t}\right] + \frac{FV}{(1 + \text{YTM}/2)^{2n}}$$

where C is the annual coupon, FV is the face value, n is years to maturity, and all semi-annual cash flows are discounted at the semi-annual rate YTM/2. In Excel the function **=RATE(2n, C/2, –Price, FV) × 2** solves this numerically in one step.

The most important property of bond pricing is the **inverse relationship between price and yield**. When market interest rates rise, newly issued bonds offer higher coupons, making existing bonds with lower coupons less attractive — their prices fall until the YTM matches the new market rate. Conversely, when rates fall, existing bonds with above-market coupons become valuable, and their prices rise. This inverse relationship is mechanical, not optional: it follows directly from the discounting formula above.

WORKED EXAMPLE 1

YTM of a Premium Bond (2-Year)

A bond with a 15% annual coupon rate and $1,000 face value is currently priced at $1,200 and matures in two years. What is its YTM? Because the coupon is paid semi-annually, there are n × 2 = 4 periods and each coupon payment is 15% × $1,000 ÷ 2 = $75.

Step	Calculation	Result
1	Number of periods = 2 × 2	n = 4
2	Semi-annual coupon = 15% × $1,000 ÷ 2	$75.00
3	=RATE(4, 75, –1200, 1000) in Excel	Semi-annual r = 2.17%
4	Annualize: 2.17% × 2	YTM = 4.34% per year

Because the bond trades at a **premium** (price $1,200 > face $1,000), the YTM (4.34%) is lower than the coupon rate (15%). Investors are paying extra for the high coupon stream, which compresses their yield. The capital loss at maturity — receiving $1,000 for a bond purchased at $1,200 — offsets part of the generous coupon income.

WORKED EXAMPLE 2

YTM of a Deep-Premium Bond (10-Year)

The same bond — 15% coupon, $1,000 face value — now matures in 10 years and trades at $1,500. How does the longer maturity change the YTM?

Step	Calculation	Result
1	Number of periods = 10 × 2	n = 20
2	Semi-annual coupon = 15% × $1,000 ÷ 2	$75.00
3	=RATE(20, 75, –1500, 1000) in Excel	Semi-annual r = 3.83%
4	Annualize: 3.83% × 2	YTM = 7.66% per year

The longer maturity raises the YTM from 4.34% to 7.66%, even though both bonds have the same coupon. With 10 years to maturity, the capital loss of $500 (= $1,500 – $1,000) is spread over 20 semi-annual periods — a smaller drag per period — but the investor also locks up capital for much longer. The market demands a higher yield to compensate for that duration risk. Both bonds remain at a premium (price > face), so both YTMs remain below the 15% coupon rate.

3. The Risk-Free Rate and the Federal Reserve

Every bond yield in practice is composed of two pieces: a **risk-free rate** and a premium for credit risk. Before understanding credit spreads, we need to understand where the risk-free rate comes from.

The **Federal Reserve** sets the **Federal Funds Rate** — the overnight rate at which commercial banks lend reserve balances to one another. Although the Fed does not directly control long-term Treasury yields, its policy rate anchors the entire yield curve. When the Fed raises rates, banks' funding costs rise, which pushes up rates across maturities and asset classes. The 2022–2023 tightening cycle — from 0.25% to 5.50%, the fastest increase in 40 years — illustrated this transmission with unusual clarity: mortgage rates doubled in less than 18 months, corporate bond yields jumped, and bond prices fell sharply.

Rate	Who sets it	Typical maturity	Effect
Federal Funds Rate	Federal Reserve	Overnight	Anchor for all other rates
T-Bill yield	Market, anchored by Fed	1m – 1yr	Short-term benchmark
10yr T-Bond yield	Market expectations	10 years	Mortgage & corporate benchmark
30yr T-Bond yield	Market expectations	30 years	Insurance & pension benchmark

US Treasury bonds are regarded as **risk-free** because they are backed by the full faith and credit of the US government, which has never defaulted on its debt. Treasury yields therefore represent the minimum return any investor requires — the compensation purely for giving up liquidity and taking on inflation risk, with no default premium. Every other borrower in the world must offer more than the Treasury to attract investors.

4. Credit Spreads and Default Risk

The fundamental equation of fixed income is:

Yield on risky bond = Risk-free rate + Credit spread

The **credit spread** (also called the yield spread) compensates investors for three distinct risks: the **probability of default** (PD), the **loss given default** (LGD, the fraction of principal that is not recovered if the issuer fails), and an **illiquidity premium** (because corporate bonds trade less frequently than Treasuries). The spread is quoted in **basis points** (bp), where 100 bp = 1 percentage point.

A practical illustration: the 30-year mortgage rate is approximately equal to the 10-year Treasury yield plus a spread of 250–300 bp. When the Fed raised rates in 2022 and the 10-year Treasury yield rose 50 bp, mortgage rates rose by almost exactly 50 bp as well — the spread was roughly constant, so the Treasury move passed through dollar-for-dollar to homebuyers. This is the **rate transmission mechanism** in action.

Borrower	Rating	Approx. Spread (bp)	Approx. Yield (2024)
US Government	Risk-free	0	~4.3%
Apple Inc. (AAPL)	AA+	~84	~5.1%
Investment-grade corp	BBB	~120	~5.5%
High-yield corp	BB / B	~370+	~8%+
Distressed debt	CCC	1,000+	14%+

CASE STUDY

Evergrande: When Bond Markets Led the News

China Evergrande Group was once the country's largest property developer, carrying more than **$300 billion in liabilities** at its peak. Its collapse is a textbook demonstration that bond markets price default risk in real time, well before rating agencies act or official announcements are made.

By mid-2021, Evergrande's bonds were trading at 20–30 cents on the dollar, implying credit spreads of more than 2,000 bp. A spread of that magnitude communicates one thing clearly: investors believe default is not a tail risk but a near-certainty, and they expect to recover only a fraction of their principal.

Period	Spread (approx.)	Market Signal
2019	~500 bp	Elevated concern; risk priced in
2020	~1,000 bp	Serious alarm; distressed territory
Early 2021	~2,000 bp	Near-certain default priced in
December 2021	Bond price: 10–20¢ on $1	Official default declared

The key lesson is that **bond markets are forward-looking**. Rating agencies had still not downgraded Evergrande to junk when its spreads had already exceeded 2,000 bp. Investors who relied solely on ratings were caught flat-footed; investors who watched spreads had ample warning. This is why practitioners say

that real-time credit spreads are a better signal of issuer health than any single rating letter.

EXERCISE 1 · MORTGAGE RATE · 2 POINTS

Q1 (1 pt): Use the formula Mortgage Rate = T-Bond Rate + Credit Spread to estimate today's 30-year mortgage rate. Look up the current 10-year Treasury yield and add the typical mortgage spread (~250–300 bp). Express your answer as a percentage to two decimal places.

Q2 (1 pt): The Federal Reserve hikes the federal funds rate by 50 bp (0.50%). Assume the 10-year T-bond yield rises by 50 bp and the credit spread remains constant. What is the new mortgage rate?

Key insight: When the Fed tightens, the increase passes through almost one-for-one to mortgage borrowers — the spread is sticky in the short run, so the full rate hike shows up as a higher monthly payment.

5. Credit Rating Agencies

The three dominant credit rating agencies — **Moody's**, **Standard & Poor's (S&P)**, and **Fitch** — evaluate the creditworthiness of bond issuers and assign a letter-grade rating that summarizes their assessment of default probability and recovery prospects.

Moody's	S&P; / Fitch	Category	Meaning
Aaa	AAA	Investment grade	Highest quality
Aa1–Aa3	AA+–AA–	Investment grade	Very high quality
A1–A3	A+–A–	Investment grade	High quality
Baa1–Baa3	BBB+–BBB–	Investment grade	Minimum IG (the "IG floor")
Ba1–Ba3	BB+–BB–	Speculative	Below investment grade ("junk")
B1–B3	B+–B–	Speculative	High default risk
Caa–C	CCC–C	Distressed	Very high default risk
D	D	Default	Already in default

The dividing line between **investment grade** (Baa3/BBB– and above) and **speculative grade** (Ba1/BB+ and below) is commercially critical. Many institutional investors — pension funds, insurance companies, and certain mutual funds — are legally or contractually restricted to holding only investment-grade securities. A downgrade below BBB– triggers mandatory sales ("fallen angel" selling), which can accelerate a borrower's distress.

The Evergrande case, however, highlights the main **limitation** of ratings: they tend to lag the market. Agencies' methodologies rely on financial statements and management meetings, which update slowly. Credit spreads, by contrast, incorporate all publicly available information continuously. When Evergrande's

bonds were already trading at distressed levels, its official rating was still investment grade — a dangerous divergence that practitioners learn to watch for.

The **risk-return spectrum** in fixed income is as follows: US Treasuries yield approximately 4.3% with zero default risk; investment-grade corporates yield roughly 5.5%, adding about 120 bp for credit risk; high-yield bonds yield 8% or more, compensating for materially higher default probabilities; and distressed or near-default bonds may yield 14% or more, reflecting the expectation that significant principal may be lost. Higher yield is not synonymous with better return — it is compensation for taking on additional default risk, liquidity risk, and ratings volatility.

6. Application: Apple's 30-Year Bond

We now apply the YTM formula and the credit spread framework to a real bond. Apple Inc. (ticker AAPL, rated AA+ by S&P) issued 30-year bonds in 2024. The specific bond — identified in markets as *AAPL 4.45 2054* — matures in 2054 and pays a 4.45% annual coupon semi-annually. We also need a Treasury benchmark to compute the credit spread.

Parameter	Apple Bond (AAPL 4.45 2054)	Treasury Benchmark (30yr)
Face value	$100	$100
Coupon rate	4.45%/yr	4.50%/yr
Payment frequency	Semi-annual	Semi-annual
Maturity	30 years (n = 60 periods)	20 years (n = 40 periods)
Current price	$93.779	$105.25
Semi-annual coupon	$2.225	$2.25

WORKED EXAMPLE 3

Step-by-Step: Apple YTM and Credit Spread

Step 1 — Calculate Apple's YTM using the RATE function:

Step	Calculation	Result
1	n = 30 × 2 = 60 semi-annual periods	60 periods
2	Semi-annual coupon = 4.45% × $100 ÷ 2	$2.225
3	=RATE(60, 2.225, –93.779, 100) in Excel	2.42% semi-annual
4	Annualize: 2.42% × 2	Apple YTM = 4.84%

Apple's price ($93.779) is **below par** ($100), so the bond trades at a discount. This means the YTM (4.84%) must be higher than the coupon rate (4.45%) — consistent with the price–yield inverse relationship.

Step 2 — Calculate the Treasury benchmark YTM:

Step	Calculation	Result
1	n = 20 × 2 = 40 periods	40 periods
2	Semi-annual coupon = 4.50% × $100 ÷ 2	$2.25
3	=RATE(40, 2.25, –105.25, 100) in Excel	2.05% semi-annual
4	Annualize: 2.05% × 2	Treasury YTM = 4.10%

Step 3 — Compute the credit spread:

Credit spread = Apple YTM – Treasury YTM = 4.84% – 4.10% = 0.74% ≈ 74 bp

A spread of roughly 74–84 bp (market quotes vary slightly depending on the exact benchmark chosen) is consistent with Apple's AA+ rating. It is an extremely narrow spread by corporate bond standards — the market is essentially treating Apple as nearly default-proof, charging only a small premium above the US government's borrowing cost.

EXERCISE 2 · APPLE BOND · 3 POINTS

Q1 (1 pt): Using =RATE(60, 2.225, –93.779, 100) × 2, compute Apple's YTM to two decimal places. **Answer: ≈ 4.84%.**

Q2 (1 pt): Using =RATE(40, 2.25, –105.25, 100) × 2, compute the Treasury benchmark YTM. **Answer: ≈ 4.10%.**

Q3 (1 pt): Compute the credit spread in percentage points. Convert it to basis points (multiply by 100). **Answer: 0.74% ≈ 74 bp.** At ~74–84 bp over Treasuries, Apple's AA+ rating is reflected in a spread that sits near the bottom of the investment-grade range — close to risk-free for a corporate issuer.

Summary

1. **YTM is the key pricing metric.**
It is the single discount rate that equates the present value of all coupons and the face value to the current price. Price and yield move in opposite directions — always.

2. **Maturity amplifies price sensitivity.**
Longer-maturity bonds are more sensitive to yield changes because distant cash flows are discounted more heavily. A 10-year bond's YTM was 3.32 percentage points higher than that of the 2-year bond in Example 2.

3 **The Fed anchors all rates.**
The Federal Funds Rate is the overnight benchmark from which every other interest rate in the economy is derived. When the Fed moves, mortgages, corporate bond yields, and savings rates all adjust.

4 **T-bonds define the risk-free baseline.**
US Treasury yields are the minimum return any investor accepts. Every other rate is quoted as a spread above Treasuries — the wider the spread, the riskier the issuer.

5 **Credit spreads price default risk in real time.**
Spreads incorporate probability of default, loss given default, and illiquidity. Evergrande's spreads reached 2,000 bp months before the official default — bond markets lead, ratings lag.

6 **Apple trades near risk-free.**
At ~74–84 bp over the 30-year Treasury, Apple's credit spread reflects its AA+ rating and near-zero perceived default risk — a concrete illustration of how the credit spread framework maps market perception to a single number.